TLE Differencing

Post Hoc Uncertainty Estimation

A.C.

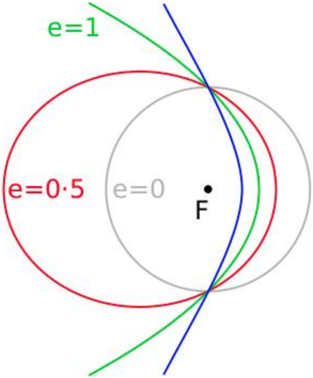
May 26, 2026

Outline

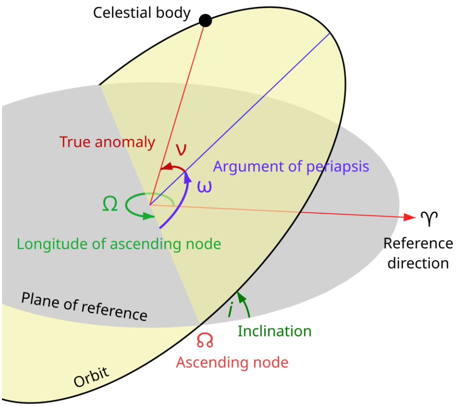
Conic Perfection

$$\ddot{R} = \frac{\mu}{r^2}$$

Conic Perfection

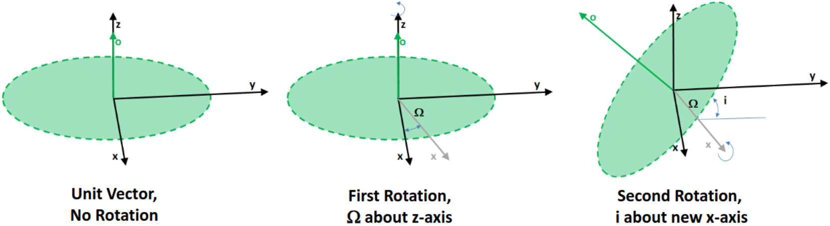


Kepler Elements



Element	Function
i	Orientation
Ω	
ω	
e	Shape
a	
ν	Position

Kepler and Euler



Kepler elements result in busy diagrams, but can be quite handy.
E.g., i and Ω directly define the direction of orbital momentum.

Perturbation: Non-Sphericity

$$\ddot{R} = \ddot{R}_{\text{2B}} + \ddot{R}_{\text{NS}}$$

- $\ddot{R}_{\text{2B}}$ uses point-mass equations
  - Works for points
  - Works for spheres (shell theorem)
- Earth is neither
  - Centrifugal forces (order kilometers)
  - Tidal forces (order meters)
- Notable for causing nodal precession.

Perturbation: Drag

$$\ddot{R} = \ddot{R}_{\text{2B}} + \ddot{R}_{\text{NS}} + \ddot{R}_{\text{D}}$$

- Drag equation: $F_{\text{D}} = \frac{1}{2}\rho v^2 C_{\text{D}} A$
- Scales by
  - Object shape and orientation (C_{D}, A).
  - Square of object velocity v^2.
  - Atmospheric density ρ, drops exponentially with altitude.
- LEO objects (altitude < 2000 km) are low and fast
  - experience non-negligible drag
- Bonus: non-periodic and non-conservative.

Perturbation: et cetera

We've only scratched the surface.

- Third bodies (esp. sun, moon)
- Tidal forces (solid and not)
- Solar radiation pressure (esp. HAMR)
- Knocke rediffusion
- Relativistic corrections
  - Schwarzschild
  - Lense-Thirring
  - De Sitter

Osculating Elements

Perturbed orbits will *not* be ellipses. Can we pretend?

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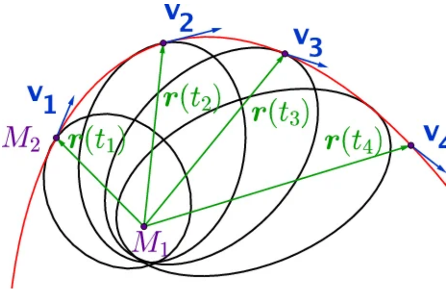

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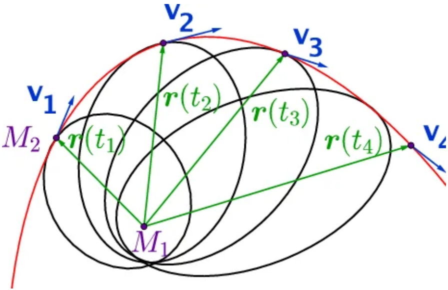
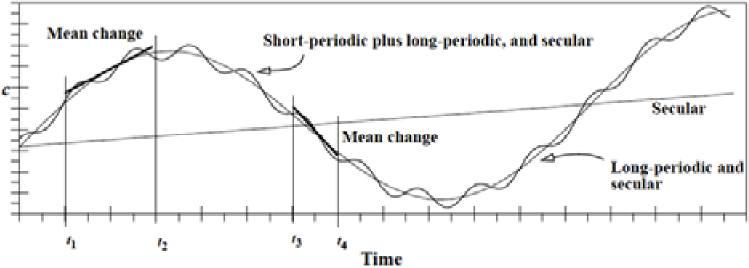


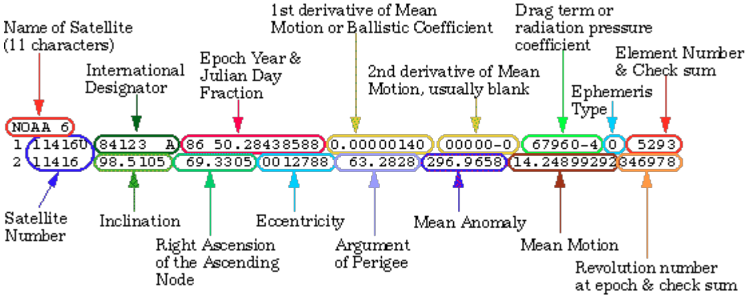
Yes, but. . .

Mean Elements



Could we "average" the high-frequency perturbations away somehow?

SGP4 and TLEs



Propagation Error

$$\ddot{R}_{\mathrm{SGP4}} = \ddot{R}_{\mathrm{2B}} + \ddot{R}_{\mathrm{NS}} + \ddot{R}_{\mathrm{D}} + \ddot{R}_{\mathrm{sun}} + \ddot{R}_{\mathrm{moon}}$$

- SGP4 catches only the biggest perturbations.
- Liable to accumulate kilometers of error within an orbital period.

Error at Epoch

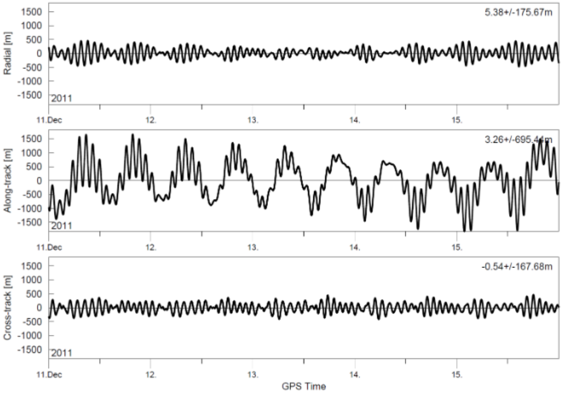


Figure 1 Orbit Error of the Fitted TLE during OD Period (5d Fit)

Precision

All models are wrong. Can we get estimates on uncertainty? Perhaps a covariance matrix?

Precision

Outline

Differencing

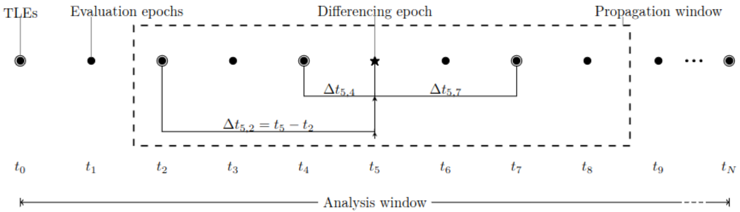
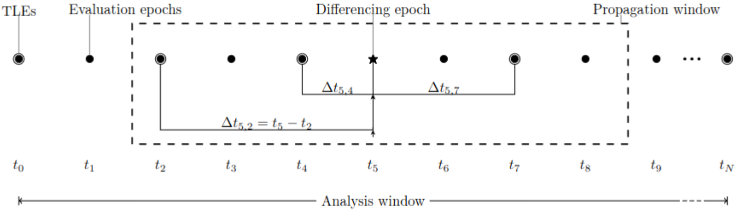


Figure 1: Schematic overview of TLE differencing method.

1. Select an RSO's TLE history to examine.

Differencing



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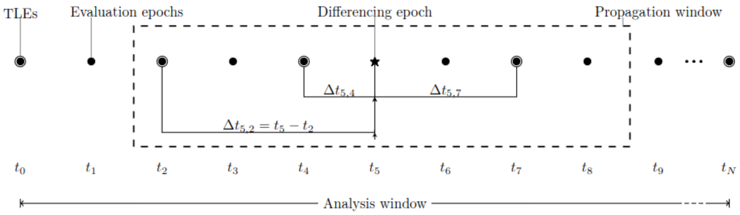
2. Assume residuals ϵ are consistently distributed over an *Analysis Window*.

Differencing



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3. Choose a *Propagation Window* over which SGP4 should be "good enough".

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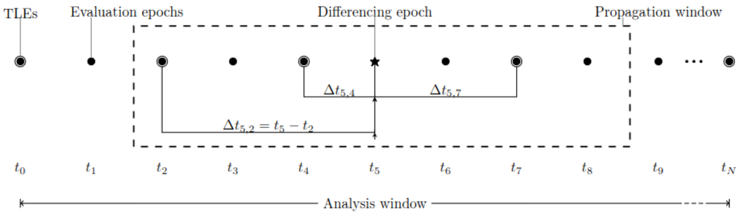

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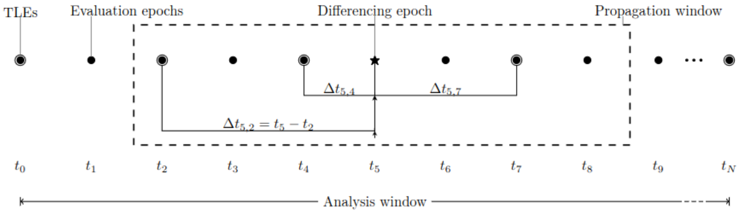

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5. Fit function $\hat{\epsilon}(\Delta t)$ to gathered samples.

Differencing Recapitulated

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Fit to Variance

Recall: standard OLS model

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Prediction error spread:

$$\mathrm{Var}(y_{\mathrm{new}} - x_{\mathrm{new}}\beta) = \sigma^2 + \sigma^2 x_{\mathrm{new}} (X^T X)^{-1} x_{\mathrm{new}}^T$$

Example Results

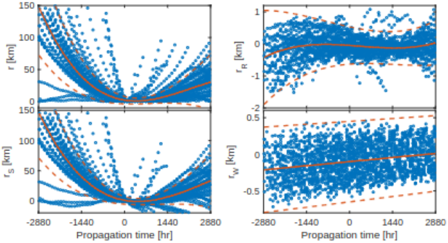
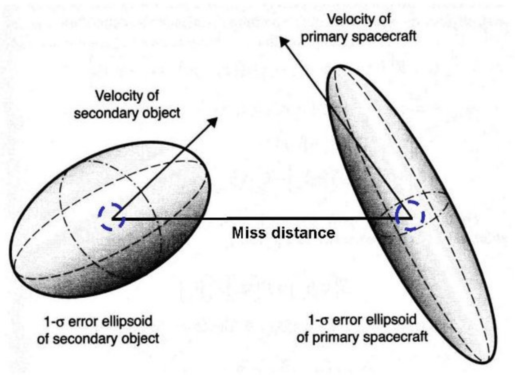


Figure 4: Robust regression of the position errors and (3σ) uncertainty bounds decomposed in the RSW frame.

Reference Frame



Residuals naturally oriented along satellite motion (RIC frame).

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Why inverse square? No theoretical justification provided.

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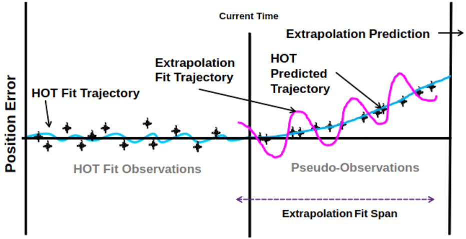

Figure 1: Extrapolation DC fitting approach diagram.

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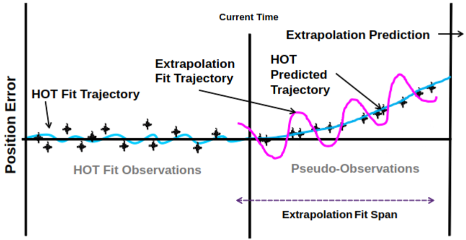

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In fact TLEs may not be most accurate at epoch!

Error Model

Recall: final step of differencing protocol was to fit an error estimating function $\hat{\epsilon}(\Delta t)$.

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Polynomials? Rationals? ANNs?

Just interpolating, data is “well-behaved” – low order polynomial is fine. Plus, interacts well with OLS.

Outlier Handling: Rejection Multipliers

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Commonly effective, but $r_{\max}$ is a nuisance hyper-parameter.

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WLSQ and Bi-square weight function are topics for another day, but in summary:

1. WLSQ minimizes a weighted L2 norm.
2. Bi-square weight gives more weight to well-modeled obs and less to poorly modeled obs.

GNSS

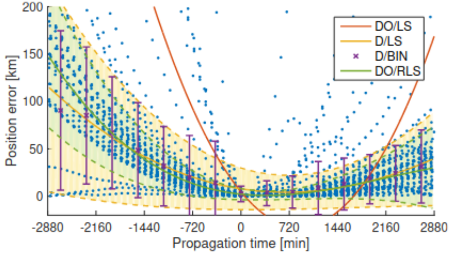
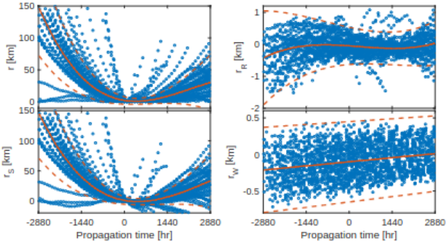


Figure 3: Benchmark of different methods for estimating the error and (3σ) uncertainty of the TLE propagation errors of Period 1a.

GNSS readings (blue) provide truth baseline for evaluation.

In-Track



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As expected, in-track error dominates.

Back To Covariances

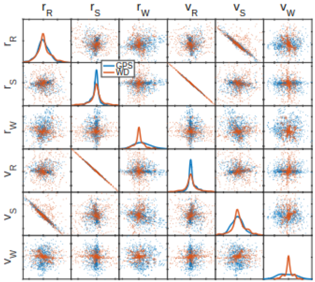

Figure 9: Correlation estimation using weighted differences compared to GPS of classic TLEs.

Independent polyfits do not covariance compute. But combined with correlations from raw sample statistics. . .

References and Further Reading

[1] Glenn E Peterson, Robert G Gist, and Daniel L Oltrogge. "Covariance generation for space objects using public data". In: *Proceedings of the 11 th Annual AAS/AIAA Space Flight Mechanics Meeting, Santa Barbara, CA*. 2001, pp. 201–214.

[2] Béatrice Deguine et al. "Covariance modelling in satellite collision risk activities". In: *AIAA/AAS Astrodynamics Specialist Conference and Exhibit*. 2002, p. 4631.

[3] Saika Aida and Michael Kirschner. "Accuracy assessment of SGP4 orbit information conversion into osculating elements". In: (2013).

[4] MD Hejduk et al. "A catalogue-wide implementation of general perturbations orbit determination extrapolated from higher order orbital theory solutions". In: *Proceedings of the 23rd AAS/AIAA space flight mechanics meeting*. 2013, pp. 619–632.

[5] Jacco Geul, Erwin Mooij, and Ron Noomen. "TLE uncertainty estimation using robust weighted differencing". In: *Advances in Space Research* 59.10 (2017), pp. 2522–2535.

Self Link

The source for this presentation is hosted at https://github.com/alan-christopher/edu/tle-differencing.